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Abstract

This paper covers the process of building and fine-tuning a line-following robot using an IR/LDR sensor, from setting up the motors and sensors to optimizing the final code. Along the way, we ran into some challenges—like wiring issues and mismatched motor headers—but tackled them one by one. The robot was tweaked to handle sharp turns smoothly, thanks to an improved lost-line recovery system.

Introduction

The goal of this project was to design and build a working line-following robot that could autonomously navigate a predefined path. The focus was on motor control, sensor integration, and fine-tuning movement algorithms to keep things smooth and efficient. Along the way, we ran into a few hardware and software hiccups—like miswired motors and sensor grounding issues—but tackled them step by step to get the robot running optimally. The final results showed solid line tracking and better lost-line recovery, allowing the robot to handle sharp turns more accurately. This report breaks down the theory, methods, results, and key takeaways from the experiment.

Background Theory

A line-following robot is a type of autonomous mobile robot that detects and follows a set path—usually a black line on a white surface—using infrared (IR) sensors. The way it works is pretty straightforward: black surfaces absorb more infrared light, while white surfaces reflect it, allowing the sensors to figure out the robot's position relative to the track.

Motor control plays a key role in keeping the robot moving smoothly and in the right direction. The line-following robot we designed uses a differential drive system, where two independently controlled wheels adjust speed to steer. To make navigation more precise, a control algorithm is used.

Previous research has shown that fine-tuning motor behavior is essential, as no two motors perform exactly the same. By calibrating motor offsets, straight-line tracking and overall accuracy can be improved. Another important aspect is **lost-line recovery**—adjusting the robot's response to sharp turns or sudden track deviations. Increasing the aggressiveness of sensor-based corrections helps the robot quickly find the line again when it veers off course.

Methodology

1. Motor Connection and Testing

- The first motor's pin was connected to ground, the second pin to 5V, and its rotation was tested. The same process was followed for the second motor.
- **Issue Encountered:** The motor headers were swapped, meaning the left motor was controlled by the right header and vice versa.
- **Resolution:** The logical mapping was corrected in the control code.

2. Line-Following Sensor Setup

- Initially, seven bundled wires were found. However, due to mismatched color coding, they had to be replaced for clarity and safety.
- A test code was flashed onto the IR sensors to verify proper operation.
- **Issue Encountered:** Incorrect ground wiring resulted in all sensor values being consistently low.
- **Resolution:** The wiring was rechecked and corrected, ensuring proper ground connections.

3. Motor Characterization

- Offset variables were introduced to balance left and right motor speeds.
- After straight-line tests, it was determined that the right motor needed an offset of -41 PWM at 180 full speed to achieve balance.

4. Fine-Tuning of Line-Following Code

- Variables responsible for 'lost-line recovery' were adjusted to allow more aggressive turns on sharp corners.
- The motor characterization data was incorporated to optimize tracking accuracy and smoothness.

Motor characterisation:

Motor characterization in an Arduino-controlled vehicle typically involves measuring and compensating for inherent discrepancies in motor performance. In this project, each motor's PWM (Pulse Width Modulation) signal was adjusted to correct deviations from a straight path. Initially, the robot veered to the left by varying distances, indicating that the right motor was operating more efficiently or at a higher speed. By reducing the PWM value supplied to the right motor (e.g., decreasing it by 30 or 41 from the initial setting), the imbalance was corrected, resulting in the robot traveling in a straight line.

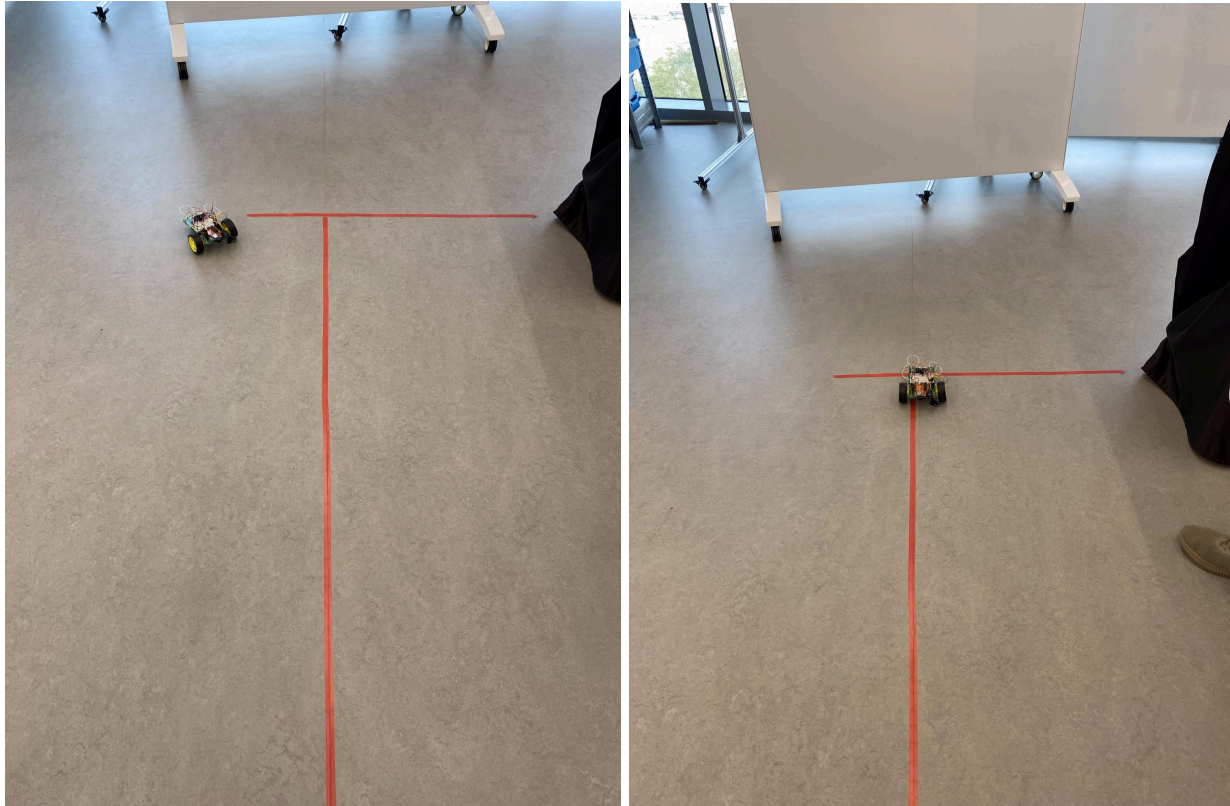
Such adjustments reflect the need to account for manufacturing tolerances, frictional differences, and load variations. Characterizing each motor allows the control algorithm to ensure consistent, predictable movement. This is a fundamental step in robotics projects, where precise motion control is critical to achieving reliable navigation and task execution.

Appendix A

<u>Initial Motor Value Left</u>	<u>Initial Motor Value Right</u>	<u>Accuracy of straight line before adjustments</u>	<u>Test Adjustments made</u>	<u>Accuracy of straight line after adjustments</u>
255	255	19cm sway left	-15 PWM Right motor	4 cm left
180	180	23cm Sway left	-25 PWM Right motor	1 cm left
130	130	Never reached end	-45 PWM Right motor	2 cm right

<u>Initial Motor Value Left</u>	<u>Initial Motor Value Right</u>	<u>Accuracy of straight line before adjustments</u>	<u>Adjustments made to be straight</u>	<u>Accuracy of straight line after adjustments</u>
255	255	19cm sway left	-30 PWM Right motor	0cm
180	180	23cm Sway left	-30 PWM Right motor	0cm
130	130	Never reached end	-41 PWM Right motor	0cm

Appendix B

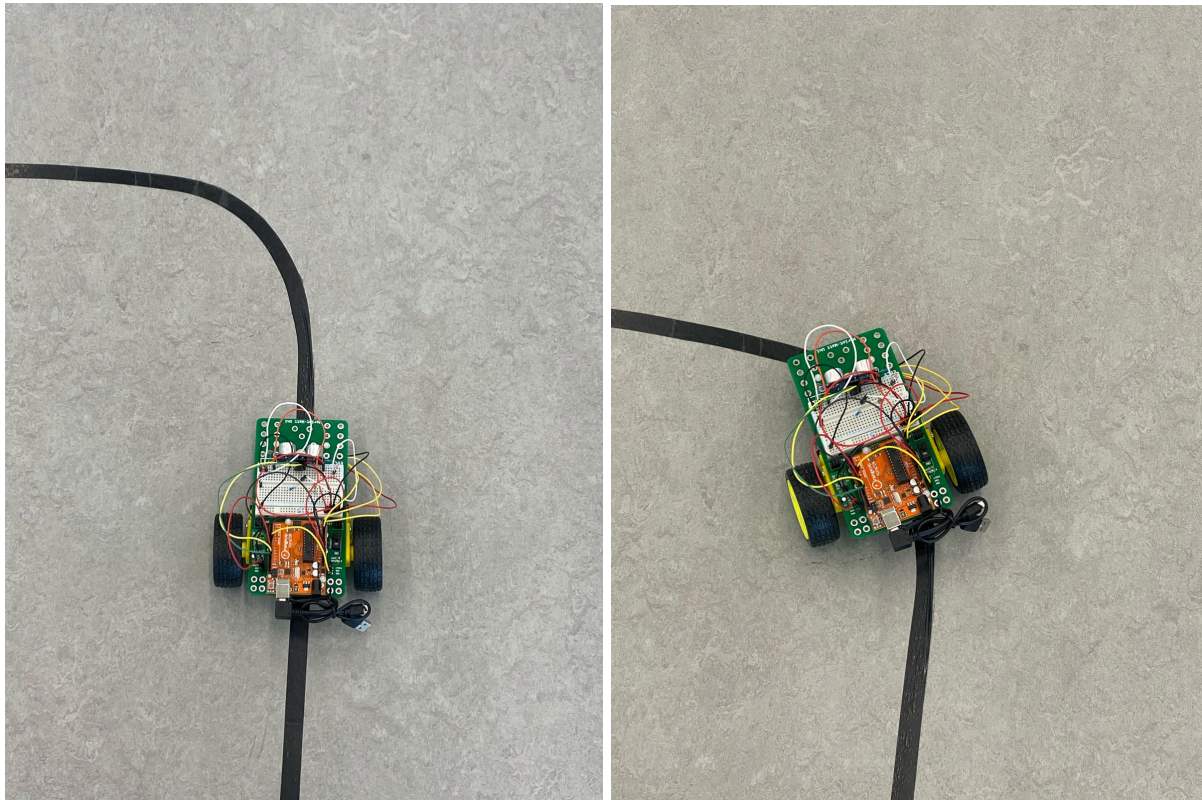


Sensor Characterisation:

The robot is equipped with both infrared (IR) sensors and light-dependent resistor (LDR) sensors. The IR sensors detect the high-contrast line against the background, while the LDR sensors monitor ambient light conditions to help compensate for environmental variability. Over multiple runs spanning 2–3 weeks, data were collected under consistent conditions to assess key parameters such as sensor voltage levels, response times, and output consistency. Analysis revealed that the IR sensors reliably detect the line due to their high sensitivity and consistency, while the LDR sensors effectively track ambient light changes, thereby supporting the overall sensor calibration and navigation algorithm.

While the baseline performance met the fundamental requirements for effective navigation, the analysis also identified areas where further enhancements could be beneficial. Statistical evaluations, including measures of mean values, variance, and signal-to-noise ratios, suggest that improvements—such as adaptive thresholding or additional filtering techniques—might further refine sensor accuracy. These enhancements could be particularly useful in environments with variable lighting or surface inconsistencies. In conclusion, although no modifications were made to the original code, the sensor characterization confirms that the default settings provide a dependable foundation for the robot's operation, while also outlining potential avenues for future development to optimize performance further.

Appendix C



Ultrasonic sensor

Originally, the robot lacked any collision detection sensors, which resulted in frequent crashes when obstacles blocked its path while it was solely focused on line following. To overcome this critical design flaw, ultrasonic sensors were integrated into the robot's system to provide accurate distance measurements and effective collision avoidance. Rigorous testing was carried out, as detailed in Appendix D, where the sensitivity of the ultrasonic sensors was varied to gauge their accuracy. The results demonstrated that the predicted distances were extremely close to the actual measurements, thereby significantly improving the robot's performance and reliability.

The table below summarizes the test results, illustrating the close alignment between the expected and actual distance measurements:

Appendix D

Effective range test:

<u>Distance</u>	<u>Reading 1 (CM)</u>	<u>Reading 2 (CM)</u>	<u>Reading 3 (CM)</u>	<u>Average (CM)</u>
1	101	101	101	101
2	202	202	202	202
3	304	305	306	305
4	406	407	406	406
5	0	0	0	0

Material affection test (setup = 1m Distance)

<u>Material</u>	<u>Reading 1 (CM)</u>	<u>Reading 2 (CM)</u>	<u>Reading 3 (CM)</u>	<u>Average (CM)</u>
White wall	101	101	101	101
Glass	101	101	101	101
Mirror	102	102	102	102

Reflection angled test(Setup = Distance 20cm)

<u>Angle</u>	<u>Reading 1 (CM)</u>	<u>Reading 2 (CM)</u>	<u>Reading 3 (CM)</u>	<u>Average (CM)</u>
90	20	20	20	20
45	24	18	21	21
15	0	0	0	0

Three tests were performed to evaluate the ultrasonic sensor's performance. The effective range test showed accurate readings up to four meters, confirming the sensor's reliable detection at moderate distances. The material effect test revealed minimal variation when detecting white walls, glass, and mirrors, indicating that surface composition has negligible impact on measurement accuracy. Finally, the reflection angle test demonstrated that angles greater than 15 degrees lead to reduced detection accuracy, making the sensor less reliable when objects are not perpendicular. Overall, these results strongly suggest that the ultrasonic sensor is robust for most applications, with slight limitations at extreme angles.

Results

- The robot successfully navigated the test track after implementing the necessary hardware corrections and fine-tuning.
- The lost-line recovery feature significantly improved performance on sharp turns.
- Motor characterization adjustments ensured straight-line movement with minimal drift.
- Figures illustrating motor speed adjustment can be found in Appendix A.
- Appendix B presents the initial test conducted without PWM adjustments, followed by the subsequent corrections made.
- Appendix C contains images that illustrate the correlation between LDR tracking and the line.

Discussion

Throughout the development process, various technical challenges were encountered and resolved. The miswiring of motors and IR sensors initially hindered performance but was rectified through debugging. The implementation of motor characterization proved essential in overcoming hardware inconsistencies. Fine-tuning the control variables allowed the robot to make sharper turns while maintaining stability.

Conclusion and Recommendations

This project demonstrated the importance of systematic testing and debugging in the development of an autonomous robot. The results indicate that effective motor characterization and fine-tuned sensor integration significantly improve navigation accuracy. Future improvements could include implementing PID control for more precise navigation and using higher-resolution sensors for improved line detection. Additionally, testing on different track surfaces would help assess adaptability under varying conditions.